

SIMATS ENGINEERING
C02_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1522
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COURSE OUTCOME COVERED

C02: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)	Total Marks: 50
Question Count: 5	

Code of Conduct

I Brijesh.T(192521391) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____



Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92% , memory 88% , storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1- Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

ANSWERS:

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter

A hypervisor is a software layer that helps us create and run virtual machines on a physical computer. There are mainly two types of hypervisors: Type-1 and Type-2.

A Type-1 hypervisor runs directly on the physical hardware of the server. It does not need a normal operating system in between. Examples include VMware ESXi and Microsoft Hyper-V.

A Type-2 hypervisor runs on top of an operating system. For example, VirtualBox can be installed on Windows or Linux and then used to create virtual machines.

Difference between Type-1 and Type-2

Type-1 Hypervisor	Type-2 Hypervisor
Runs directly on hardware	Runs on a host operating system
Gives better performance	Performance is comparatively lower
More suitable for cloud datacenter	More suitable for personal use and testing
Better control over hardware resources	Depends on the host operating system
More secure for large-scale system	Has an extra security layer because of the host OS
Can manage many VMs efficiently	Usually used for fewer VMs

Performance

Performance is very important when a cloud datacenter is hosting mission-critical applications. Type-1 hypervisors have better performance because they communicate more directly with the hardware. There is no extra operating system taking resources between the hypervisor and the hardware.

In Type-2, the request has to pass through the guest VM, the hypervisor, and then the host operating system. Because of this extra layer, some performance is lost.

Security

Security is another important factor. In Type-1, there is no general-purpose host operating system that can become an additional attack point. This makes the design more secure when properly configured.

In Type-2, the host operating system is very important. If the host operating system is attacked, the virtual machines may also be at risk.

Final choice

For a mission-critical cloud datacenter, I would choose a Type-1 hypervisor. It gives better performance, stronger security, and better management of a large number of virtual machines. Type-2 is useful for learning and testing, but Type-1 is more suitable for an important cloud environment.

2. Isolation of Finance, Healthcare and Student Workloads

Suppose a cloud provider wants to run applications related to finance, healthcare, and student portals on the same physical server. These applications contain different types of information and should not be allowed to freely access each other.

Virtualization helps solve this problem by dividing one physical server into multiple virtual machines.

The hypervisor controls the resources given to each VM. This prevents one VM from directly using the resources belonging to another VM. Because of this separation, the three workloads can run on the same physical machine without mixing their data.

How virtualization provides isolation

First, each VM gets its own virtual memory. This means a program running inside one VM cannot normally read the memory of another VM.

Second, the hypervisor controls CPU access. A VM cannot simply take over the CPU resources assigned to another VM.

Two major attack surfaces

1. Hypervisor

The hypervisor is a very important security point because it controls all the VMs. If an attacker finds a serious weakness in the hypervisor, they may try to escape from one VM and reach the host system or other VMs. This type of attack is dangerous because many workloads may be affected at the same time.

2. Virtual network

The virtual network is another attack surface. If the network is not properly separated, an attacker who gains access to one VM may try to communicate with another VM. For example, a compromised student portal should not be able to freely reach a finance database.

Protection methods

To improve security, the cloud provider should use proper network segmentation, strong access control, secure hypervisor settings, regular security updates, monitoring, and firewall rules.

Conclusion

Virtualization makes it possible to run different workloads on the same hardware while keeping them separated. However, virtualization does not automatically make a system completely secure. The hypervisor and virtual network must be protected carefully

because they are important attack surfaces.

3. Study of VM Host Utilization Snapshot

The given VM host has the following values:

- CPU utilization = 92%
- Memory utilization = 88%
- Storage I/O latency = 35 ms
- Average VM boot time = 170 seconds

These values show that the server is under heavy load.

Memory problem

The memory usage is 88%, which is also high.

Virtual machines require memory to run their operating systems and applications. When the available memory becomes low, the system may have difficulty handling new workloads. In some situations, the host may use disk space as temporary memory, which is much slower than RAM.

Because of this, high memory usage can also contribute to poor VM performance.

Storage problem

The storage I/O latency is 35 ms. This means storage requests are taking a noticeable amount of time to complete.

Virtual machines perform many read and write operations during startup. They also continuously use storage while applications are running.

If the storage system is slow, the VM may take more time to start. This is probably one of the reasons for the 170-second average VM boot time.

Corrective actions

This process is often called VM migration or workload balancing.

The provider can also:

- Add more CPU capacity.
- Increase RAM.
- Move workloads to faster SSD or NVMe storage.
- Reduce unnecessary applications running inside VMs.
- Spread VMs across multiple hosts.
- Monitor the server regularly to avoid overload.

Conclusion

The main bottlenecks are CPU, memory, and storage. CPU usage of 92% and memory usage of 88% show that the host is overloaded, while 35 ms storage latency is slowing down

disk operations. Together, these problems are causing the high VM boot time of 170 seconds.

4. How Software Defined Datacenter Improves Agility

A Software Defined Datacenter (SDDC) is a datacenter in which computing, storage, and networking resources are controlled mainly through software.

In a traditional datacenter, many tasks depend on physical hardware. For example, adding a new server may require buying the hardware, installing the system, connecting cables, and configuring it manually.

This process takes more time.

In an SDDC, many of these tasks can be done using virtualization and software-based management. This makes the datacenter more flexible and faster.

Compute virtualization

Compute virtualization divides the resources of one physical server into multiple virtual machines.

For example, one physical server may run several VMs for different applications.

If the company suddenly needs more computing power, new VMs can be created without purchasing a completely new server immediately.

This makes resource allocation faster and improves hardware usage.

Storage virtualization

Storage virtualization combines storage resources from different physical devices and presents them as a common storage system.

Instead of managing each storage device separately, the administrator can manage storage from a central system.

If one application needs more storage, additional storage can be allocated through software. This is much more flexible than manually installing a new storage device every time.

Network virtualization

Network virtualization allows administrators to create virtual networks, virtual switches, and other network services through software.

For example, separate virtual networks can be created for different departments or applications.

Network settings can also be changed more quickly without making large physical changes to the network hardware.

Conclusion

SDDC improves agility because compute, storage, and network resources are virtualized

and controlled using software. It makes the datacenter easier to scale, faster to manage, and more flexible than a traditional hardware-based datacenter.

5. Multi-Cloud Federation and Identity Mismatch

A multi-cloud environment means an organization uses services from more than one cloud provider.

For example, an organization may use one provider for storage, another for applications, and another for databases.

A common problem in this type of environment is identity mismatch.

Different cloud providers may use different usernames, roles, permissions, and authentication systems.

For example, a user may have the role of administrator in one cloud, but the second cloud may use a different role name or permission structure. Because of this difference, the user's access may not work correctly.

Problems caused by identity mismatch

The main problems are:

- Users may need different accounts for different clouds.
- Managing many accounts becomes difficult.
- Permissions may not match between providers.
- The wrong permissions may be given to a user.
- Security management becomes more complicated.

It can also create a privacy problem because user information may have to be stored separately by different providers.

Identity federation

A good solution is identity federation.

In identity federation, a trusted organization or system called an Identity Provider (IdP) verifies the user.

After the user is verified, the cloud services can trust the identity information provided by the identity system.

This means the user does not need to create separate accounts for every cloud provider.

Secure identity design

A secure design should include the following:

1. Central identity management

Use one trusted identity system to manage users and their authentication.

2. Multi-factor authentication

Users should use more than just a password. A second verification method can provide

additional security.

3. Least privilege

Users should only receive the permissions they actually need. For example, a student user should not receive administrator-level access.

4. Short-lived tokens

Instead of giving permanent access credentials, temporary tokens can be used. This reduces the risk if a token is stolen.

5. Protect user privacy

Only the required user information should be shared with cloud providers. There is no need to send unnecessary personal information.

6. Secure communication

Identity information and authentication messages should be protected while being transferred between systems.

Conclusion

The multi-cloud project failed because the cloud providers did not have a common way of handling identities and permissions. Identity federation can solve this problem by allowing different cloud providers to trust a common identity system. A secure design using MFA, least privilege, limited data sharing, and temporary tokens can improve both security and privacy.